LAMPROS KONSTANTELLOS

Electrical & Computer Engineer | Dipl.-Ing. (Integrated Master's)

Athens, Greece (willing to relocate) | +30 6949362354 | lampros.konstantellos@gmail.com | LinkedIn

PROFESSIONAL SUMMARY

Electrical and Computer Engineer with research and consulting experience spanning PV, wind, BESS, and EV charging systems. Master's thesis at TUM on PV and V2G integration in island grids (3rd Best Paper, IEEE PESS 2025). Skilled in energy system modelling, techno-economic analysis, and ML-based forecasting. Two peer-reviewed publications. Seeking research or advanced engineering roles in energy systems and flexibility.

PROFESSIONAL EXPERIENCE

LDK Consultants SA, Greece — May 2025 – Present

Renewable Energy Consultant (Hybrid)

- Perform technical due diligence and feasibility studies for PV, wind, and BESS projects across IFI-financed (EBRD, EIB, World Bank) and private-sector mandates.
- Conduct energy yield assessments, CAPEX/OPEX modelling, and financial analysis (NPV, IRR, BCR, payback) to support investment decisions.
- Prepare technical and financial proposals for international tenders; review EPC & O&M contracts, permitting, and interconnection documentation.

Technical University of Munich, Germany — Mar 2024 – Feb 2025

Master's Thesis Researcher (Hybrid)

- Co-supervised by Prof. T. Hamacher (TUM) and Prof. G. Konstantopoulos (Univ. of Patras); research led to IEEE PESS 2025 3rd Best Paper Award.
- Built a real-time HIL model (Typhoon HIL) of the Kastellorizo island MV grid integrating a 0.5 MW PV plant and two 250 kW bidirectional V2G chargers.
- Demonstrated V2G integration improved frequency stability by 20–22%, reduced diesel output by up to 33%, and cut CO_2 emissions by ~360 kg per 3-hour cycle.

Fraunhofer ISE, Freiburg, Germany — Oct 2024 – Jan 2025

Project Intern — EV Charging System Analysis (On-site)

- Evaluated commercial EV charger performance under solar-optimized scenarios using Power Hardware-in-the-Loop (BMWK-funded "Wallbox-Inspektion" project).
- Developed a BiGRU neural network in Python to forecast 5-hour charging profiles from 5-minute data; achieved <5% energy prediction error.
- Quantified financial and emissions impact using 2023 German electricity prices and grid emission factors. Results published at EVS38 (Gothenburg, 2025).

University of Patras, Greece — Jun 2022 – Dec 2025

Research Assistant (Hybrid)

- Managed communication, dissemination, and reporting for three EU Horizon projects.
- Contributed to an IoT-based water monitoring system integrating Arduino hardware with Modbus/MQTT protocols.

Technical University of Munich, Germany — Jul 2023 – Sep 2023

Research Intern — Energy System Optimization (On-site)

- Applied 'urbs' LP optimization and 'pyGRETA' renewable assessment tools to design decarbonization pathways for a medium-voltage network.
- Proposed integration of 6.72 MW ground-mount PV, 2.41 MW rooftop PV, 1.12 MW wind, and 51 MWh battery storage by 2052, reducing costs by €500/MWh.

University of Patras, Greece — Oct 2023 – May 2024

Laboratory Teaching Assistant (On-site)

- Guided student groups in Power System Analysis and Control & Stability lab courses; received positive feedback for clarity and hands-on methods.

EDUCATION

University of Patras, Greece — Oct 2019 – Jul 2025

Diploma – Integrated Master's (300 ECTS), Electrical & Computer Engineering

- Grade: 7.54/10.0 (Thesis: 10.0/10.0) | Ranked 7th in cohort (2019–2020 intake).
- Thesis: “Integration of PV Power and V2G Technology in Electric Power Systems of Non-Interconnected Islands: A Case Study of Kastellorizo” — in collaboration with TUM.

PEER-REVIEWED PUBLICATIONS

L. Konstantellos, A. Mohapatra, T. Kavvathas, G. Konstantopoulos, T. Hamacher, “Integration of PV and V2G Technology in an Island Grid: A Real-Time Simulation Study of Kastellorizo,” *IEEE PESS 2025*, pp. 31–36, doi: 10.30420/566656006. **3rd Best Paper Award.**

L. Konstantellos, Z. Kamaci, D. Pekmezci, B. Köpfer, “Financial Impact Analysis of EV Charging Behavior with RNN Model and Validation Against Real-World Data,” *EVS38*, Gothenburg, Sweden, Jun 2025.

TECHNICAL SKILLS

Programming & ML: Python (Pandas, NumPy, TensorFlow/Keras, Scikit-learn, Matplotlib), MATLAB, Simulink
Energy Tools: PVsyst, Typhoon HIL, AutoCAD, urbs, pyGRETA
Other: LaTeX, MS Office Suite, VS Code, Jupyter Notebook, Git

LANGUAGES

Greek (Native) | English C2 – Certificate of Proficiency, Univ. of Michigan | German B1 – Goethe-Zertifikat B1 (preparing B2, May 2026)